Receipted Recursive Recovery of Negative Quantum-Method Results

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Abstract

Negative results in computational research are often recorded only as prose, discarded after a method changes, or reopened without an explicit rule for what changed. We study an executable alternative in this work: prospectively freeze the question, comparison set, gates, and honest terminals; give the strongest admissible donor method first right of refusal; retain nulls, refutations, and donor absorptions as first-class records; type the failure and authority state that permits reopening; and bind result-bearing computations to replayable receipts. The contribution is methodological rather than a new quantum algorithm. Across one end-to-end programme, this discipline produces all of the outcomes it is supposed to permit. Several plausible R6 candidate families are exactly absorbed or refuted before a bounded structural residual is isolated; the N1–N4 recovery studies contain both donor closures and exact-synthetic positives; protected data remain sealed when a frozen gate does not license access; and capability or orchestration success never self-grants scientific authority. We use these cases to separate three properties that are often conflated: a result can be reproducible without being scientifically valid, scientifically negative without being an execution failure, and methodologically informative without being a novel method. The evidence is a one-programme case study, not a statistical evaluation of research-methodology effectiveness. Claims owned by the companion TARE, dual-instrument, and typed-state papers are treated only as outcomes generated under the common discipline. A submission-date literature review narrows the positioning residual to the prospective typed-terminal/authority and receipt-binding contract; no exhaustive priority claim is made.

Keywords: negative results; research provenance; reproducibility; autonomous research; prospective protocols; donor comparison.

1 Introduction

A negative result has at least three possible meanings: the candidate mechanism is wrong, the current representation/search family is inadequate, or the computation could not establish either conclusion. Research software commonly collapses these states. A failed run may be treated as negative evidence; a new idea may silently replace the comparison set; a donor-equivalent method may be described as new; or an old negative may be reopened without recording which assumption changed.

This paper studies a narrower object than a general theory of scientific discovery: an executable discipline for preserving and recursively revisiting negative computational results. The discipline is built around five constraints. Outcome rules are frozen before result access. Donor methods receive first right of refusal. Exact claims carry witnesses or independently checkable referees. Failure and authority are typed so a capability result cannot self-authorize a scientific conclusion. Finally, the record is replayable enough that a later reader can reconstruct what was known, what was attempted, and why a lane stopped.

This programme provides a stress test because its history contains the outcomes that a non-gamed method should retain: donor absorptions, exact refutations, bounded positives, mixed results, and unresolved/cannot-check states. The strongest mathematical result of that programme is reported separately; the dual-instrument benchmark is reported separately; the partial-knowledge mechanism studies belong to separate work. Here they matter only as evidence that the common methodology did not force them into one preferred terminal.

The central claim is therefore descriptive and methodological: this receipt-first negative-recovery workflow was instantiated end-to-end and behaved in a non-monotone, outcome-sensitive way under prospectively frozen rules. We do not estimate how often such a workflow improves science, and we do not claim that receipt completeness is a substitute for evidentiary quality.

2 Methods

2.1 Prospective freeze

Each result-bearing lane binds its subject, admissible information, candidates/baselines, endpoints, gates, tie-breaks, and terminal vocabulary before outcome inspection. Later corrections are logged rather than retroactively changing a gate. This makes a refutation or donor tie an admissible result rather than an incentive to move the goalposts.

2.2 Donor first right of refusal

A candidate receives scientific residual credit only after the strongest faithfully composable existing mechanism has been run or otherwise discharged under the same admitted information. A tie or donor win closes the candidate claim. Donor absorption is therefore a positive methodological terminal even though it yields zero candidate novelty credit.

2.3 Typed failure and authority

The workflow separates scientific state from orchestration state. Missing capabilities, failed tools, and malformed receipts are control conditions, not evidence against a hypothesis. Scientific/candidate receipts cannot by themselves grant novelty, adoption, promotion, merge, or global-stop authority. Reopening is tied to typed changes in assumptions, interfaces, representations, or evidence rather than to generic dissatisfaction with a negative result.

2.4 Exactness and hostile controls

Where a claim is finite and exact, the programme uses exhaustive enumeration, proof-carrying dynamic programming, independent witness reconstruction, or complete local machine checks. Protocols predeclare hostile controls designed to make an attractive shortcut fail. If the control does not bite where required, the world/lane is invalid rather than counted as a positive.

2.5 Receipt and replay contract

Result-bearing computations serialize canonical inputs, outputs, identifiers, and authority boundaries. Content-addressed or hash-bound receipts permit deterministic replay or independent recomputation. The contract establishes attribution and reproducibility; it does not establish the truth of a scientific inference without the corresponding design/referee.

2.6 Case-study selection

We analyze the complete R6 saturation/recovery arc and the N1–N4 closure families because they were defined by the programme before this paper synthesis and include heterogeneous terminals. The paper does not select only successful recoveries. Companion papers own the domain-specific claims.

3 Results: behavior of the methodology

3.1 R6: repeated refusal before a bounded residual

The R6 chain first produces donor absorption and several exact negatives: progressively richer joint compiler families add no value beyond stronger donor constructions on the frozen open subjects. A support-dominance audit then converts repeated collapse into a structural explanation while explicitly leaving a Tag-coupling gap. The first closure claim fails at that declared gap; its repair is later refuted by a converse coupling; support-two enlargement restores exactness on the tested domains and is subsequently upgraded to an all-n theorem in a separate companion work.

For the present paper, the important observation is procedural. The workflow reports the early candidate families as donor-equivalent or refuted rather than promoting them as separate methods, yet it does not stop searching merely because several negatives accumulated. Each next question is licensed by a specific residual left by the preceding result.

3.2 N1–N4: recovery is not a one-directional success story

The N-lane studies exercise the same discipline on different partial-knowledge and synthesis questions. N1 contains a bounded typed-failure-state benefit whose allocation policy is exactly closed by an ideal value-of-information donor. N2 includes a prospectively supported crossover predictor that is subsequently donor-absorbed on the original world, leaving only a narrower misspecification result. N3's registered synthesis families survive their donor comparisons on the stated exact-synthetic domains. N4's six partial-knowledge families support typed/scoped state mechanisms against matched-information controls while retaining designed negative regimes.

These outcomes cannot be pooled into a meaningful single success fraction: they test different questions, and donor absorption is intentionally a scientifically different terminal from candidate support. Their common methodological value is that the record preserves the distinction.

3.3 Protected and orchestration boundaries

The R6 prospective gate records an honest 'R6_EARNED = NO' outcome while the protected stretched-N2 discriminator remains sealed. The harness/campaign records also keep host or capability failures outside the scientific evidence channel. This is evidence that the implementation can refuse to turn unavailable execution or unreleased protected data into a favorable scientific terminal.

3.4 Replayability

The publication packet indexes protocols, result receipts, and replay checks across the case studies. Deterministic lanes reproduce canonical outputs; exact claims include independent or alternate reconstruction where specified. Replay discrepancies, if any, are a publication blocker rather than something to be averaged away.

4 Discussion

The case studies suggest that recursive recovery is most useful when it is treated as a rule-governed change of question, not as persistence. Reopening after a donor closure without a new residual would be search inflation. Refusing to reopen after a representation or access boundary changes would treat an old negative as globally permanent. Typed state and prospectively declared reopen conditions provide the bridge between those extremes.

The R6 history illustrates a second distinction: saturation and explanation are not the same. Several exact negatives can establish that a family adds no value on a scope, but a later dominance theorem or counterexample is needed to explain why. A receipts-first archive makes that transition possible because the negative sequence remains available to be compared with the later structural account.

Third, the method separates provenance from authority. Reproducible evidence can still be badly designed, use a weak donor, or support only a narrow inference. Conversely, an honest ‘CANNOT_CHECK‘ can be methodologically correct even though it offers no scientific answer. A reviewer should therefore evaluate the scientific design of each companion result independently of this paper’s provenance claim.

The unresolved research question is comparative: does this discipline improve the reliability, efficiency, or calibration of research programmes relative to other workflows? The current evidence does not answer it. Doing so would require a prospectively defined multi-programme evaluation with matched tasks and independently adjudicated outcomes.

5 Related work and claim boundary

The workflow combines established research practices rather than claiming them individually. Preregistration formalizes the separation between hypotheses/analyses defined before outcomes and post-outcome explanations [1]. Scientific-computing guidance emphasizes version control, testing, automation, and recoverable workflows [2]; data-provenance research formalizes how derived outputs remain traceable to source records [3]. Tool-using autonomous scientific systems demonstrate that language-model agents can plan and execute multi-step research activities [4], while a current AutoResearch survey emphasizes evidence preservation, reproducibility, provenance, validation, and accountable closure as open system-level concerns [5]. None of these generic ingredients is claimed as new here.

Failure memory and trial-to-behavior harnesses. Wang’s negative-knowledge memory layer treats failed attempts as a shared asset: a curator converts an attempt into a bounded typed record and downstream agents must explicitly use or reject relevant records before proposing a successor [6]. Sibyl-AutoResearch moves closer to the workflow studied here by running bounded trials, preserving positive and negative outcomes, and auditing whether trial signals alter later research or harness behavior [7]. Its reported conversion events and recovered-failure registry directly establish that useful trial history can be made inspectable and can change later action; they also explicitly stop short of a comparative performance claim.

Those results remove any residual novelty credit for reusable failure memory, bounded trial preservation, or the general idea that failures should change later behavior. The narrower object studied here is a different contract: honest terminal categories and gates are frozen prospectively; the state that permits a closed route to reopen is typed alongside the terminal; and result-bearing claims are bound to replay receipts. The contribution is the observed composition of those constraints in one programme, not a claim that other failure-aware harnesses lack scientific value.

Failure diagnosis at research-workflow scale. AutoResearchEval evaluates 100 real-world frontier-research tasks across seven domains and annotates 800 agent trajectories with a 45-pattern failure taxonomy [8]. This is stronger evidence about the breadth and recurrence of autonomous-research failure modes than the single programme reported here. Its contribution is diagnostic evaluation and artifact-level process visibility; the present paper does not compete on benchmark coverage. Instead, it asks what authority and custody state a project should retain after a bounded attempt terminates. Cross-domain comparative performance of that discipline remains untested here.

Institutional refutation and correction. Schaeffer et al. argue for a dedicated refutations-and-critiques track and distinguish refutation from reproducibility: a result can replay while the surrounding claim remains misleading, and a critique requires a claim precise enough to be refuted [9]. Those observations are prerequisites, not contributions of this paper. Their proposed remedy is institutional and social; the local contract studied here is machine-checkable project state. It does not replace publication-level correction or confer the visibility and standing of a reviewed refutation.

Submission-date residual. After this subtraction, the candidate residual is deliberately narrow: an executable one-programme case study combining prospectively frozen honest terminals, an explicit typed authority/reopen condition, and receipt-bound claim custody. The manuscript makes no 'first', 'unique', or exhaustive-nearest-work claim. A closer source should narrow this residual without changing the preserved negative, absorbed, mixed, or cannot-check terminals.

6 Limitations

The evidence comes from one programme in one repository, with many protocols authored by the same research effort that later synthesizes the methodology. This supports implementation and case-study claims, not an unbiased estimate of effectiveness. The case studies are heterogeneous and cannot be converted into an aggregate recovery rate without defining a population and sampling process that do not currently exist.

Prospective freezing reduces outcome-contingent gate changes but does not guarantee that a frozen question is well chosen. Donor first refusal is only as strong as the donor search and faithful implementation. Replay guarantees attribution/reconstruction but not construct validity, external validity, or novelty. Typed authority prevents one class of self-promotion but does not replace independent scientific review.

Several studies use exact-synthetic worlds, and some live-host actions depend on model/tool behavior that is replayed from receipts rather than re-executed identically. The methodology should therefore be tested on independently designed programmes before any broader reliability claim.

7 Conclusion

This programme demonstrates an executable way to preserve and recursively revisit negative computational research results without forcing every lane toward a positive. The method freezes outcome rules, gives donors first refusal, retains negative history, types the state that licenses reopening, and binds computations to replayable records. In the observed programme it yields donor absorptions, exact refutations, bounded positives, protected-data restraint, and honest unresolved states.

That is the paper's claim ceiling. The work does not establish that this workflow is generally superior to alternative research methodologies, nor that a receipted result is automatically good evidence. Its value is a reproducible contract for saying what was tried, what failed, why a question was reopened, and which stronger claim remains unauthorized.

8 Reproducibility

The final publication package carries a claim-to-receipt map and an anonymous review supplement. Each headline statement is associated with the protocol/result class that supports it and, where applicable, with an independent replay or reconstruction route. The non-anonymous release archive preserves the exact repository paths and commit identities; those self-identifying paths are intentionally omitted from the double-blind manuscript.

Reproduction is claim-sensitive. A byte-identical receipt validates replay; an exact scientific claim additionally requires its referee/witness contract; a donor claim requires the donor comparison; protected-custody claims require evidence that the protected reference remained unreleased. A successful tool invocation or orchestration trace is therefore never sufficient on its own to promote a scientific conclusion.

The review supplement is bounded to the evidence needed to understand the methodology claim. It does not repackage optional cross-domain successor work as if that work had been executed.

9 Ethics, safety and resource statement

The studies reported here do not involve human participants, animals, clinical data, or personal data. The relevant integrity risks are selective reporting, hidden post-outcome gate changes, accidental protected-data access, and authority laundering from a successful tool or model call into a scientific conclusion. The methodology is designed specifically to make these events visible or fail closed.

Computational cost varies by lane from inexpensive deterministic synthetic studies to large exact enumerations. Resource use is not itself evidence. Negative and null computations are retained even when they are expensive, because deleting them would distort the recovery history. No capability receipt, model output, or campaign terminal can by itself authorize novelty, publication, deployment, or scientific truth.

Author contribution and AI assistance. Sze Chun Yiu is responsible for the methodology design, implementation, evidence interpretation, claim decisions, and final manuscript. Large-language-model assistance was used during publication closure for editorial refinement, bibliographic research, consistency checking, and package preparation. The model received no independent authority over scientific terminals; the author reviewed the retained claims and accepts responsibility for the final text and computations.

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